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In-situ monitoring method of femtosecond laser welding between glass and copper with acoustic emission

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ABSTRACT

The use of femtosecond laser welding has created new possibilities for connecting glasses and metals, revolutionizing the fabrication and assembly of advanced electronic components. However, monitoring the connection process during femtosecond laser welding is challenging due to the small heat-affected zone and brief interaction time. This study utilizes acoustic emission to monitor femtosecond laser welding and introduces the "Sum of Relative RMS" feature to correlate connection formation with acoustic emission signals in glass-copper welding. The mechanism and signal occurrence trends during welding are explained through in-situ signal monitoring and offline scanning electron microscopy characterization. Additionally, a Convolutional Neural Network is enhanced by increasing the convolutional kernel size, integrating multi-dimensional features, and incorporating prior knowledge via a customized loss function, resulting in an increase in the accuracy of successful welding detection from 89% to 95%. This research provides new insights into the monitoring of femtosecond laser glass-metal heterogeneous welding.

1. Introduction

Femtosecond laser welding has gained significant traction in fields such as microelectronic mechanical systems (MEMS) and fiber optic packaging. Traditional methods of bonding glass and metal materials, including adhesive bonding [1] and anodic bonding [2], have been explored. Katayama et al. [3] discussed various glass bonding techniques, highlighting the challenges in achieving a balance between the bonding process and the desired outcomes. In recent years, ultrafast laser welding has emerged as a novel method for joining transparent materials. Carter et al. [4] demonstrated that ultrafast laser pulses offer advantages such as a small heat-affected zone and precise processing. Lipat'eva et al. [5] employed femtosecond laser welding to join glass and metal with similar thermal expansion coefficients, with a shear strength of 26 MPa. Miyamoto et al. [6] reviewed the work on ultrashort pulse laser glass welding, discussing the phenomenon of laser-material interaction during glass welding process. Min et al. [7] discussed the effects of processing parameters on welding of glass and aluminum alloy. Hollister et al. [8] achieved continuous and spot welding of glass and aluminum using picosecond laser.

However, femtosecond laser processing of transparent materials brings many new challenges despite its significant advantages. For femtosecond laser welding, the small heat-affected zone and fast processing speed make it difficult to achieve high-precision, high-frame-rate capture of the welding process through current imaging methods, which would require unimaginable costs and cannot be realized in industrial applications. Sebastian Hecker et al. [9] used optical coherence tomography (OCT) to locate the weld of glass welding by using data evaluation methods, the positioning deviation was within the sub-nanometer range. Sebastian Hecker et al. [10] used high-speed cameras and photodetectors to detect plasma emission during glass welding, and predicted welding strength or cracking based on whether the molten body was regular during welding. Oleg B. Vorobyev et al. [11] recorded the Fiber Bragg Grating (FBG) reflection spectra of fiber before, during and after welding and during bending experiments by using an incoherent light source and a spectral analyzer to realize real-time state monitoring of fiber core. The equipment used in the above-mentioned research is costly, difficult to install, and challenging to implement in actual production processes. Therefore, there is a need to find an in-situ monitoring solution that is easier to install and cost-effective for widespread adoption.

By establishing the correlations between specific frequency bands or

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the features and signals, acoustic emission (AE) monitoring captures the vibration signals generated during the machining process, avoiding ultra-high frequency monitoring target. The evaluation of specific machining phenomena is performed based on the inherent features contained in these signals. In welding monitoring, Gu and Duley [12] investigated the physical mechanism of laser radiation conversion into heat using AE and optical emission monitoring, demonstrating that monitoring specific frequency bands determined by system parameters yielded better results. Kuo and Lu [13] utilized AE signals to assess the quality of butt welding, while Lu et al. [14] further employed hidden Markov models to classify AE signals in stainless steel lap welding. Luo et al. [15] detected weld penetration depth using AE signals and verified their feasibility for monitoring laser welding processes by correlating the plasma changes captured by coaxial cameras with AE signals. In the context of ultrashort pulse laser machining monitoring, Liu et al. focused on the characteristic frequency signals produced during the formation of defects in glass. They monitored nanosecond laser glass scribing [16] and crack propagation during the glass processing [17] using AE signals. For other metallic materials, Xie et al. [18] monitored the laser derusting effect on carbon steel surfaces through AE signals, Yang et al. [19] employed AE monitoring to detect laser-induced thermal damage, Zhong and Li [20] monitored the plasma shockwaves generated from femtosecond laser ablation of copper, iron, and aluminum using AE signals, thus validating the potential of AE technology for online monitoring in femtosecond laser machining. As for other materials, Xie et al. [21] established the correlation between AE signals and SiC machining processes.

It is crucial to understand the mechanism behind the process of direct bonding between glass and metal using femtosecond laser, by monitoring the formation of the connections between two substrates, the success of the welding can be determined. Direct connection of glass and metal using femtosecond lasers has been reported [4,22–26]. In these studies, they believe that the welding process is a high energy density laser applied to the base material, and the molten base material fills the gap between the two materials to achieve welding. Acoustic emission monitoring captures the vibration signals generated during the welding process using contact or non-contact sensors and evaluates specific phenomena based on the features inherent in the signals. When the gap between the materials is filled, the acoustic impedance changes from glass-air-copper to glass-copper, resulting in a sudden increase in signal intensity. Therefore, the acoustic emission signal has the ability to monitor the femtosecond laser welding process.

In this paper, an acoustic emission monitoring method designed for glass-copper welding is proposed. A new AE signal feature was designed to capture the characteristics of welding processes that require multiple repetitions. The correlation between AE signals and connection formation was established, allowing for in-situ monitoring of welding quality. Subsequently, a neural network model was utilized to evaluate the welding process. Additionally, the neural network was enhanced through techniques such as multidimensional data processing and incorporation of prior knowledge. This validation confirms the effectiveness of the monitoring method and provides guidance for industrial applications.

2. Methods

2.1. Experimental setup

To meet the requirements of industrial welding and simplify the process, the materials used in this study were only cleaned on the surface without any other pretreatment. To facilitate the placement of the acoustic emission sensor and ensure the consistency of the substrate material, a pure copper plate with dimensions of 50 mm × 100 mm × 1 mm was used as the substrate, and the glass material was quartz glass with dimensions of 10 mm × 10 mm × 2 mm.

The FemtoYL-20 industrial femtosecond laser was used for welding in this experiment. The laser had a central wavelength of 1030 nm, a power range of 0.06–26.6 W, a repetition frequency of 25–5000 kHz, and a beam diameter of 3.5 mm. The laser adopts an F-Theta-Ronar 1064 nm (Linos 4401-302-000-21) field mirror with a focal length of 100 mm and a focused spot of 15–25 μm, and uses a ProSeries 1 (Cambridge technology 61710PS1XY2-YH) galvanometer to achieve laser scanning.

In this experiment, to eliminate the variations in ablation strength and plasma shielding effect caused by changes in processing parameters, we set a constant welding parameter, so that the success of welding depends on the surface quality of the materials. In this experiment, successful welding is defined as no detachment of the welded interface when subjected to a perpendicular impact load after the completion of welding. The processing parameters are shown in Table 1. To obtain real-time acoustic emission signals during laser welding, a monitoring system based on acoustic emission signals was established. The system included an acoustic emission sensor, a preamplifier, and a data acquisition card, as shown in Fig. 1. The measurement range of the acoustic emission sensor (FUJICERA, 1045S) was 50–1300 kHz. To amplify the signal, a preamplifier with a gain of 40 dB was used. The signal acquisition card used was NI PXIe 5105 oscilloscope card with a maximum sampling frequency of 60 MHz. In preliminary experiments, it was found that the frequency of the signal was mainly concentrated within the range of 200 kHz. To obtain good acquisition results, a sampling frequency of 1 MHz was used. Additionally, a coupling agent (Medical Ultrasonic Coupling, Cofoe) was applied between the copper substrate and the acoustic emission sensor to improve the signal-to-noise ratio.

2.2. Principle of signal generation

Acoustic emission signals refer to the phenomenon where when an object is subjected to localized external forces, the material generates local stress concentration due to inherent defects, micro-cracks, or physical inhomogeneities, thereby causing minor local instability and producing acoustic wave signals. Laser ultrasound in industry monitoring usually uses a low power laser, in which case the laser induced acoustic signal is generated through the thermoelastic effect [27]. However, the principle of the acoustic signal generated by the laser welding process is different.

When a high-energy–density laser beam interacts with the material, a “plasma plume” consisting of a large amount of material vapor and partially ionized plasma is generated, as shown in Fig. 2. Compared with the mechanical impact of the laser beam on the material, the recoil force and thermal vibration generated by the plasma plume have a greater effect. Therefore, the pressure wave generated by the recoil force and thermal vibration is the main mechanism for acoustic emission during laser surface ablation, rather than the shock signal formed by the laser pulse itself contacting the material [15]. This is also an important reason why MHz-level acoustic emission signals can be used to monitor femtosecond laser processing processes. In the experiment of section 3.2, high energy density laser acts on copper and glass surfaces, causing intense ablation of the two materials, forming and ejecting a plasma plume. The reaction force and thermal vibration formed by this process form an acoustic emission signal, and the center frequency of this acoustic emission signal is affected by the inherent properties of the material, rather than the laser repetition frequency.

Table 1
Welding parameters.

Parameter	Value
Laser Repetition Frequency	100 kHz
Laser Power	1.06 W
Scanning Speed	20 mm/s
Path Spacing	50 μm
Number of Welding Repetitions	3

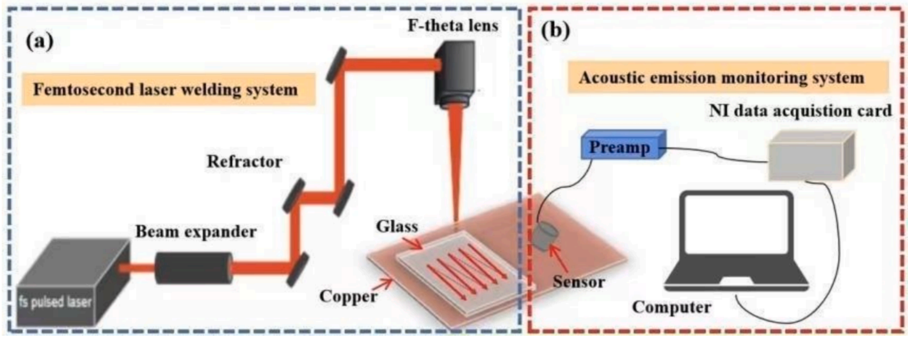

Fig. 1. Schematic of the welding monitoring system.

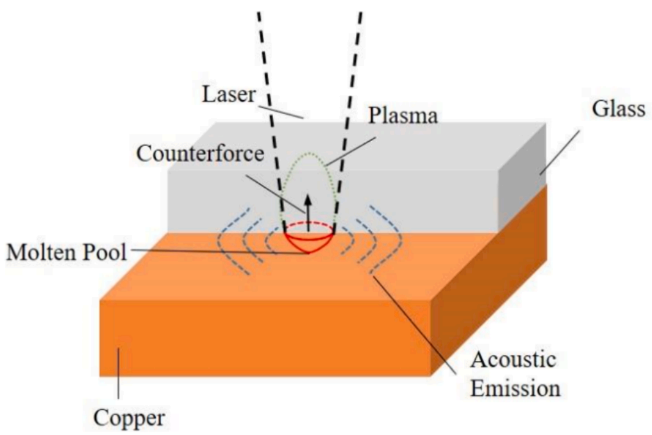

Fig. 2. Schematic of AE signal generation by laser beam.

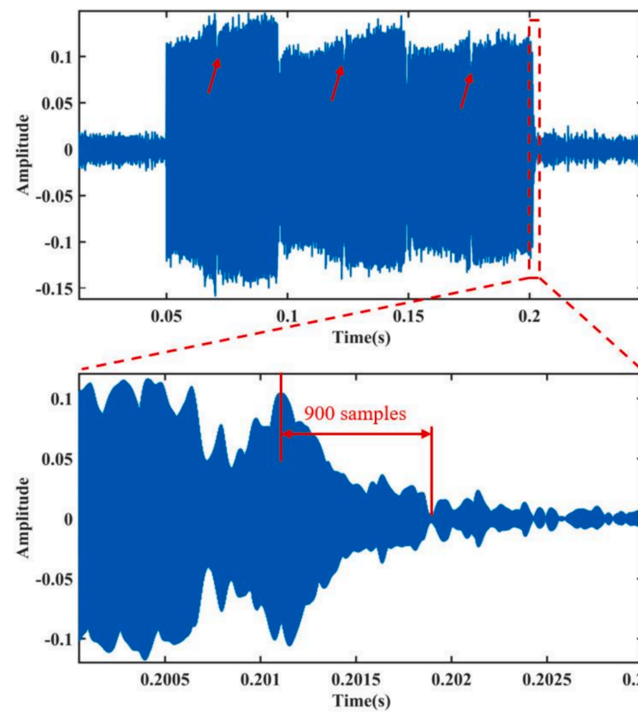

Fig. 3. Signal of laser single-line ablation repeated three times and enlarged view of signal end. The red arrows indicate signal fluctuations occurring at the same location during three repeated welds.

3. Results and discussion

3.1. Feasibility verification of acoustic emission signal

To determine whether the acoustic signals can capture the characteristics of the welding process, a preliminary experiment was conducted. The laser was focused on a copper plate, and a welding path of a 2 mm straight line was scanned three times. The corresponding acoustic emission signals are shown in Fig. 3.

To investigate the occurrence of signal overlap, we amplified the end of the processed signals. It can be observed that the signal overlaps within approximately 900 sampling points (900 μs) from the highest peak of the last pulse to the signal attenuating to the noise level. This suggests that the continuous processing signals may overlap within a maximum of 900 sampling points, while the "periodic" signal length exceeds 155,000 points. Therefore, it can be considered that the signals in the initial stage of processing do not affect the subsequent signals. Apart from the sudden signal variations caused by defects such as scratches on the sample surface (indicated by the red arrows in Fig. 3), the single-line ablation signal shows a stable overall trend. Regarding the overall trend of the three repeated processing signals, the amplitude of each segment decreases compared to the previous segment. When the laser is focused on the surface of the substrate during the first processing, the laser energy density is the highest, resulting in the most severe thermal ablation effect. As the number of laser processing increases, the defocusing effect becomes more obvious and the energy gradually decreases.

3.2. Signal analysis and feature construction

After confirming that the acoustic emission signals can capture the laser processing process, welding experiments were conducted. The processing path is illustrated in Fig. 4. The welding path is shown in Fig. 4b, the size of the welding area is 3 mm × 3 mm, the interval between adjacent welding paths is 50 μm, and the welding is repeated three times. The acoustic emission sensor is placed 5 cm away from the center of the welding area on the copper. However, because the sensor is placed manually, there is an error in its installation position, which has a certain influence on the strength of the signal. In the subsequent signal processing process, we will try to eliminate the influence by signal processing methods.

Fig. 5 shows typical processing signals and enlarged views of the weld start, second repeated weld start, and last weld end. It can be observed that in one complete welding, the signal generated by the laser on the adjacent welding path shows a certain periodic form, which indicates that the welding process is stable. However, in the enlarged view (Fig. 5(b)-5(e)), the trend of the "periodic" signals is not the same. This is due to the unevenness of the material surface and the refraction or self-focusing effect of the glass caused by different incident angles of the femtosecond laser, resulting in an inaccurate focus on the surface of the

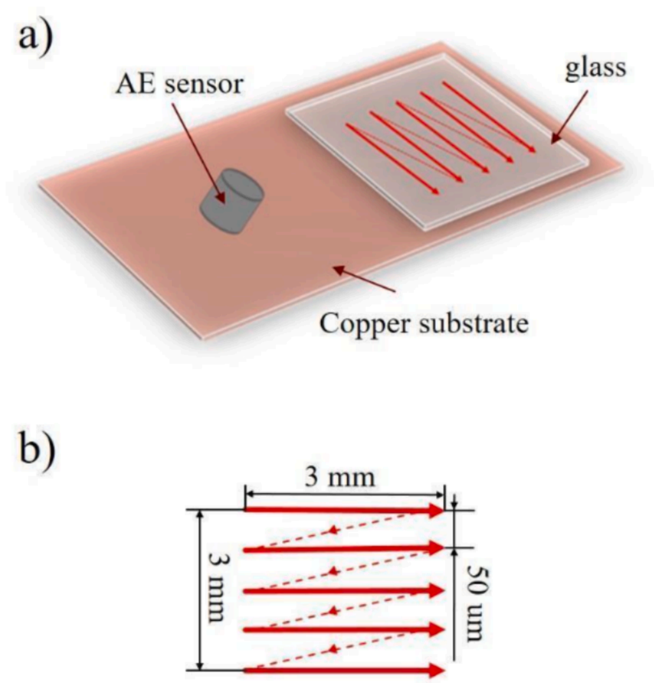

Fig. 4. Schematic diagram of glass-copper welding: (a) sample and sensor distribution, (b) welding path.

copper substrate. It can be observed in Fig. 5(a) that among the three repeated processing signals, the amplitude of the third signal is significantly higher than that of the first two signals. This contrasts with the trend depicted in Fig. 3, where the subsequent signals decrease with an increase in processing cycles. It is precisely this anomalous phenomenon that signifies the formation of connections during the processing process.

In acoustic emission monitoring, the commonly used features are AE Hits, RMS, and MARSE. Among them, AE Hits refers to the number of samples in the AE signal whose amplitude exceeds a certain threshold. This method assumes that the AE signal becomes effective for monitoring the target only when the amplitude exceeds the threshold. RMS and MARSE, on the other hand, focus on the energy characteristics within a certain time period of the signal. RMS represents the root mean square value of the signal, while MARSE represents the area under the envelope curve of the signal. These two methods consider that cumulative energy reflects the variations in the monitored target more effectively. Utilizing FFT for preliminary signal processing allows for an initial understanding of the signal's energy in specific frequency bands, facilitating the determination of the main signal frequencies. The use of RMS features enables the assessment of energy variations over specific time intervals, offering a more intuitive way to identify potential energy spikes or fluctuations during the welding process. In the present work, it is reasonable to use the RMS feature and traditional FFT spectrum for preliminary observation of the signals.

Fig. 6 shows the spectrogram of 7 groups of typical welding acoustic signals, where the red data represents successful welds and the black data represents failed welds. Under the same level of low-frequency noise, the signal energy is mainly concentrated around 200 kHz. The difference between successful and failed welding signals cannot be directly observed from the spectrogram. The RMS waterfall plot of the typical welding signals, shown in Fig. 7. It can be seen from the diagram that although the RMS value of each section has a different trend, it has an obvious periodicity. These two conventional signal analysis methods alone cannot accurately distinguish the differences in welding quality.

Under the condition of multiple repeated welds, the initial state of the subsequent weld is inevitably influenced by the previous weld. It is

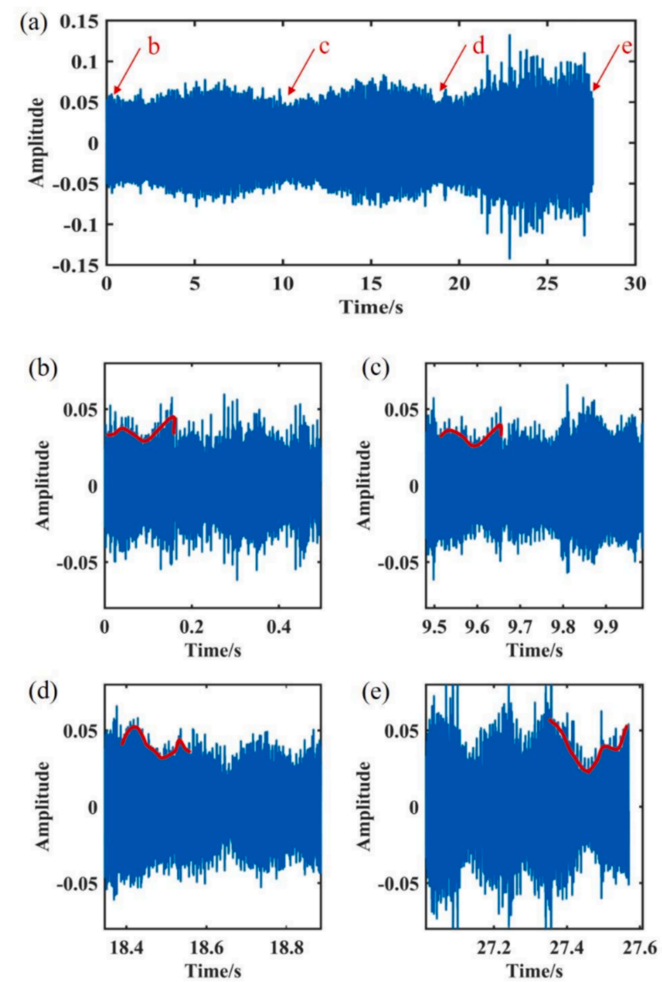

Fig. 5. Typical time waveform of processed signals and zoomed-in view of selected time intervals. (a) complete Processing Signal, (b)-(d) amplified plots of signals from different welding cycles at the same location, (e) amplified plot of welding signals at different locations. The red line represents the envelope of a processed signal.

insufficient to solely rely on RMS or FFT features for effective monitoring of repeated welds. Therefore, a feature that reflects the overall quality of multiple welding process is required.

The calculation of the RMS value of the acoustic emission signals is shown in Equation (1). In acoustics, RMS evaluates the average intensity of signal energy over a certain period. Although the first welding acoustic signal may have different intensities and trends due to the different surface morphology of the materials or the sensor position, we can still assume that the first welding signal tends to be consistent overall. By summing the relative values of the subsequent signal RMSs with respect to the RMS of the first segment, as indicated in Equation (2), the result is reflected in Table 2. Among them, Samples 1 and 7 failed to form connections at the end of welding, and Samples 3 and 4 formed successful connections but experienced detachment after a 10 cm drop impact.

$$RMS = \sqrt{\frac{\sum_{i=1}^{n} x_i^2}{n}} \tag{1}$$

$$S_{RRMS} = \sum_{i=1}^{n} \frac{RMS_{i+n} - RMS_i}{RMS_i} + \sum_{i=1}^{n} \frac{RMS_{i+2n} - RMS_i}{RMS_i} \tag{2}$$

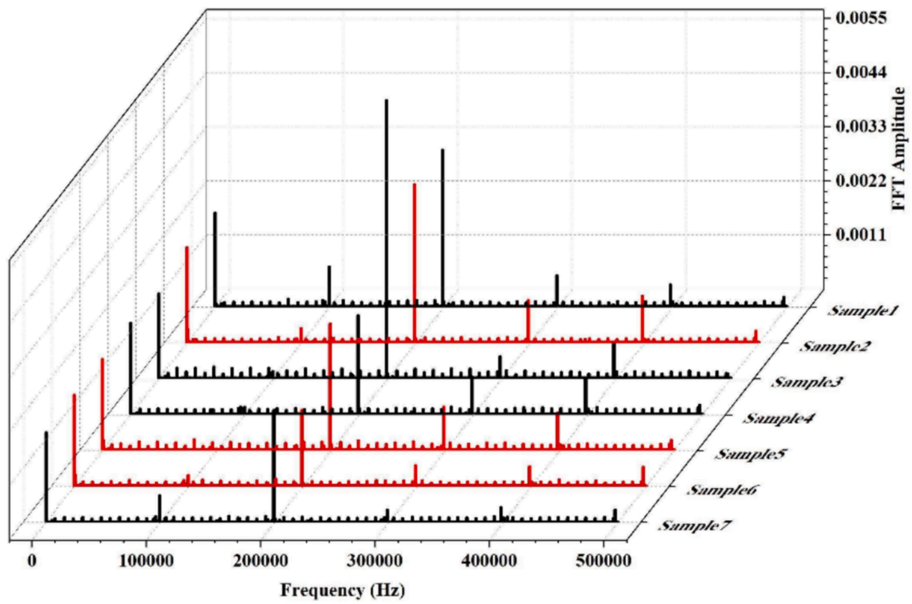

Fig. 6. Spectrogram of 7 welding signals with the same parameter. The black signal signifies welding failure, while the red signal signifies welding success. It is hard to discern the differences between successful and failed samples from the figures.

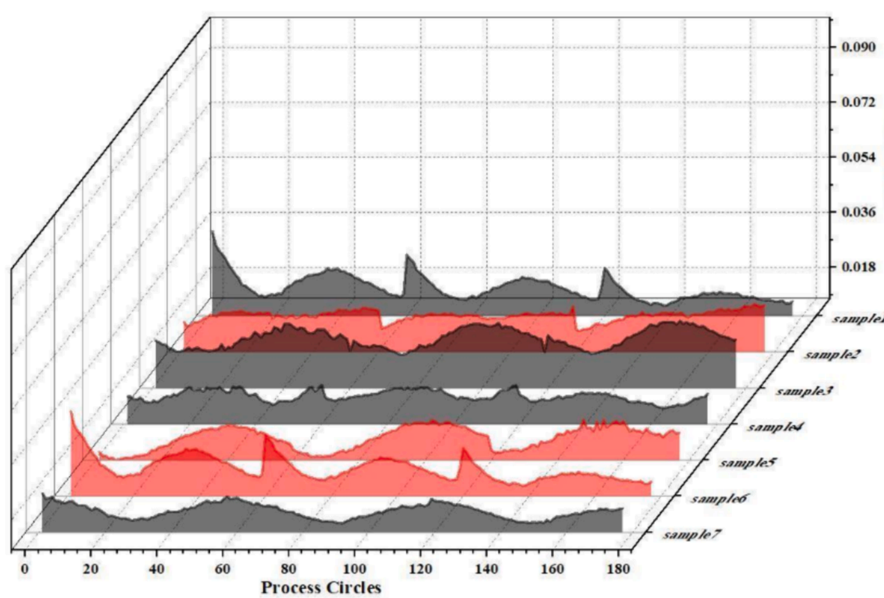

Fig. 7. RMS values of the 7 welding signals with the same parameter. The black signal signifies welding failure, while the red signal signifies welding success. It is hard to discern the differences between successful and failed samples from the figures.

Table 2
Sum of relative RMS values of the signals.

Welding results	Sample number	S_{RRMS}
Weld failure after impact	Sample1	−2.522
	Sample3	6.654
	Sample4	7.342
	Sample7	−4.399
Weld remained after impact	**Sample2**	**14.605**
	Sample5	**7.637**
	Sample6	**11.086**

Fig. 8 shows the SEM image of the welded interface after tensile failure of the welding sample. In the area outlined by the red box, it is evident that the main defects in the welding region remain unfilled. The distinction between successful and unsuccessful welds lies in the size of the filled region in the welding area. The formation of the fill will alter the propagation pattern of the acoustic signal within the sample.

Fig. 9 and Fig. 10 are SEM and EDS images of the interfaces of successful and failed weld samples respectively. In Fig. 9, the protrusion in the middle of the welding interface is the copper formed after melting and solidification, and it is also the reason for the glass-copper connection. The laser pulse is focused on the copper substrate, causing partial melting. When there is enough melted copper to fill the gap between the glass and copper or form a connection, the acoustic emission signal transitions from copper-air-glass to copper-glass due to the propagating medium. This transition results in a smaller attenuation of the signal, leading to sudden increases in the amplitude and RMS features of the signal, ultimately resulting in a larger positive Sum of

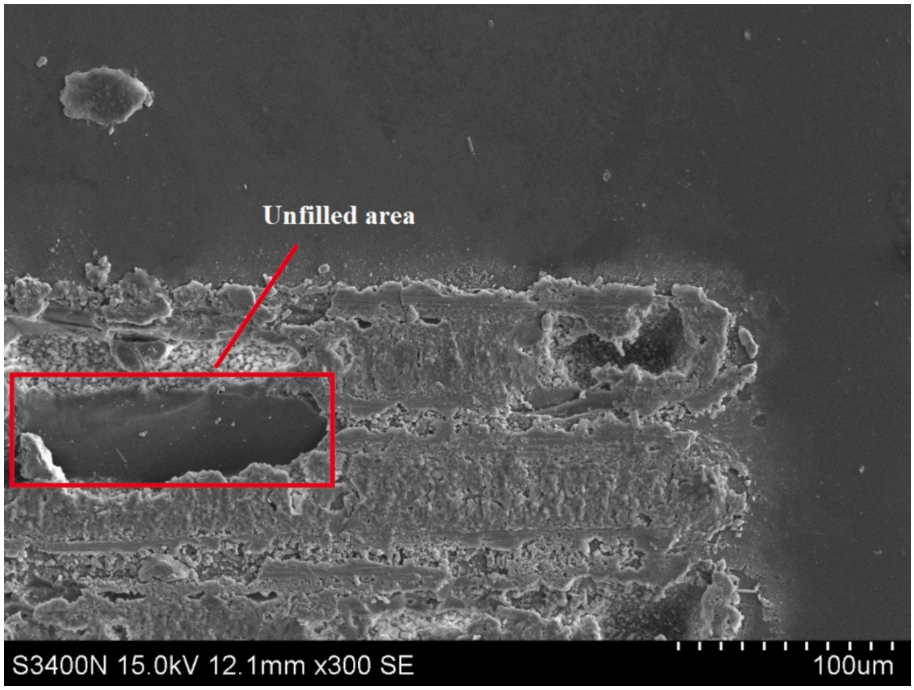

Fig. 8. SEM microscopy of the interface of the welding area.

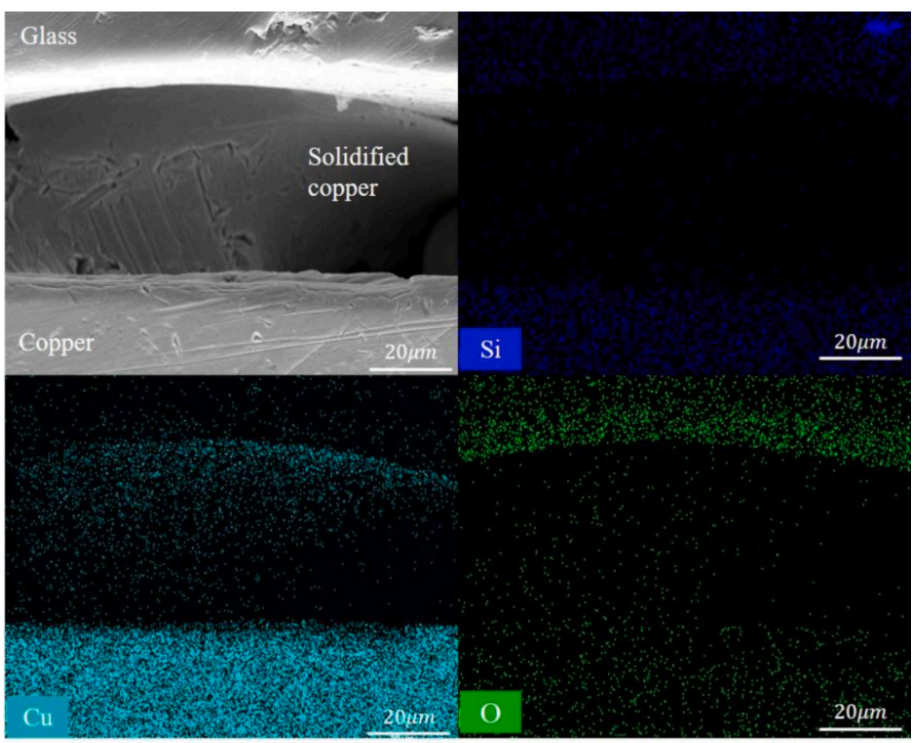

Fig. 9. SEM image and EDS element distribution map of a successfully welded sample. Copper fill the gap between glass and copper after melting and solidification.

Relative RMS value. On the other hand, in the failed weld sample (Fig. 10), it can be observed that there are defects such as pores on the glass side, which deposit surface laser energy on the glass side and cause the glass to melt. The reason for the focus deflection of the glass may be due to the refraction and self-focusing effect [28] caused by the different laser incidence angle or the error occurred during the sample assembly. Due to the presence of many unfilled areas in the welding sample, resin infiltrated into the sample gap during the preparation of the sample to take the electron microscope image, resulting in the presence of trees in the electron microscope image. The connection between the materials is insufficient or not formed, which is reflected in the acoustic signal by either the unchanged or reduced magnitude of the signal features. Consequently, the Sum of Relative RMS values exhibit smaller positive values or even negative values.

This indicates that the Sum of Relative RMS values can nearly serve as an effective indicator for connection formation and welding quality.

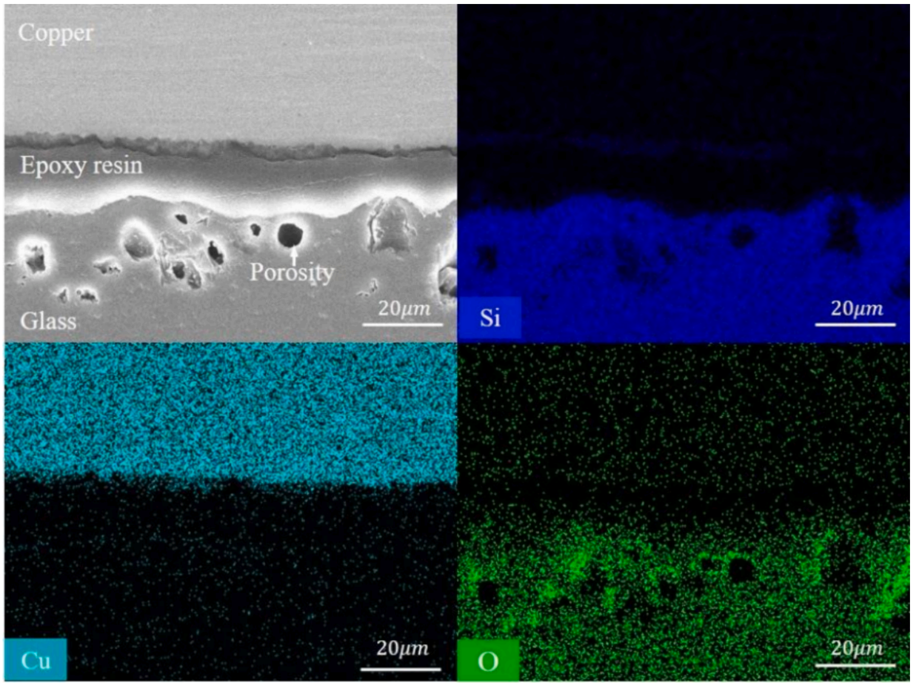

Fig. 10. SEM image and EDS element distribution map of a failed welded sample. Due to the laser focusing more towards the glass, only a small amount of molten glass adheres to the interface, resulting in insufficient melted copper for successful bonding. Consequently, defects such as pores are observed on the glass side.

However, the discriminative efficacy of SRRMS in samples 4 and 5 was not ideal. It is necessary to find a more stable evaluation standard.

3.3. Effectiveness validation and application of feature signal

Convolutional Neural Networks (CNN) is widely used in the field of monitoring. Xue et al. [29] have employed CNN to monitor various welding performance states by analyzing captured weld pool images. Alex Božič et al. [30] present a method for controlling laser power in a remote laser welding system with a convolutional neural network via a PID controller, based on optical triangulation feedback. To validate the effectiveness of the acoustic signal parameters designed in this study, CNN will be employed in the following section to evaluate and classify the experimentally collected acoustic signals.

One of the most challenging aspects in the analysis of welding process signals is the length of the signals. A common approach to dimensionality reduction is to extract features that effectively represent the target object from the signals, condensing millions of data points into a dozen or even a few features. This approach helps the model extract the essential information for the research focus, but it also sacrifices a significant amount of detailed information in the signals. This is a great waste of information for neural network models capable of fitting complex relationships. Daugela et al. [31] successfully transformed the acoustic emission signals induced by nanoindentation into time–frequency spectrograms using Continuous Wavelet Transform (CWT). By employing GoogleNet for classification, they achieved 100 % accuracy in classifying different types of plastic deformation. Similarly, Wang and Oates [32] recognized the competitive advantage of converting signals from one-dimensional to two-dimensional representations for the model.

Due to the excessive length of the signals, the signals were analysed by dividing them into segments of 15,500 sampling points each (which corresponds to one-tenth of the length of the laser scanning signal). Using cwt function of MATLAB, the signals were processed by wavelet transform. morse wavelet with center frequency of 1e6 is selected as the wavelet base, and its scale is adjusted adaptively according to the frequency of the wavelet base. The signals were transformed into spectrograms using CWT and fed into a simple CNN model to assess the feasibility of this method, as shown in Fig. 11. During the training process, there were 5,328(1776*3) images of welding success and 5,328 (1776*3) images of welding failure after dividing 6 welding sample signal and converting them to images, with a training-to-validation set ratio of 2:1 to confirm the reliability of this method. By training the model at different learning rates, the accuracy of the validation set was used to determine the feasibility of the neural network model for this task. The results, as shown in Fig. 12, indicate that when the learning rate is 1e-3, the model can only fit near the optimal solution due to the excessively large parameter update step size, resulting in a decrease in accuracy. However, when the learning rate is set to 1e-4 and 1e-5, the model's accuracy stabilizes around 80 %. This indicates that there are indeed distinguishable differences between welding success and failure signals in the spectrograms, demonstrating the feasibility of this method. However, an accuracy of 80 % is insufficient to support the industrial application of this approach. Therefore, the convolutional layers of the model were visualized to explore the problems and propose solutions.

Fig. 13 presents the visualization results of the convolutional layers of the model with a learning rate of 1e-4 during the training process, where the highest validation accuracy is achieved. The information captured by the convolutional layers can be categorized into three types, as depicted in Fig. 13(b)-13(d). In Fig. 13(b), all values are zero, indicating that these convolutional channels have failed to learn any features. Fig. 13(c) represents the texture features of the time–frequency signals, showcasing how the model learns to "observe" the time–frequency patterns without analysing them. In Fig. 13(d), the red arrows indicate frequency distributions with higher energy at specific time points, implying that the model can "decode" the time–frequency images. Among all the convolutional layers, approximately 50 % fail to learn any features, 34 % can discern texture information from the images, and only 16 % can extract more detailed time-domain features. This suggests that the model underutilizes the time-domain features present in the time–frequency images.

To address these limitations, this study proposes three targeted improvement measures: feature supplementation, enlarging the convolutional kernels, and incorporating prior knowledge.

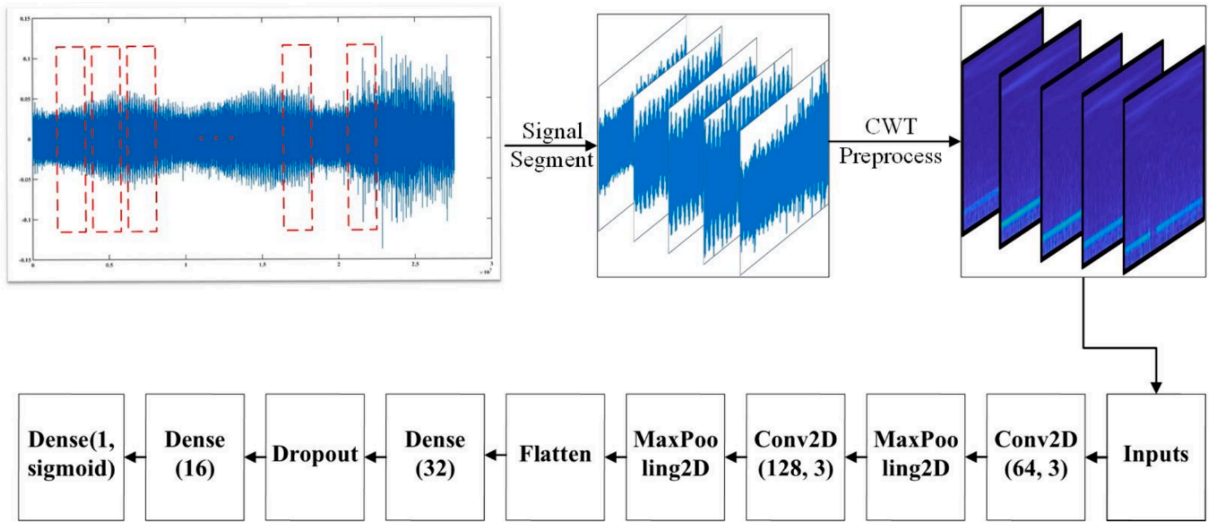

Fig. 11. Predicting welding results using CNN. The signals were segmented based on the time of a single pass and preprocessed using CWT before being fed into the neural network for classification.

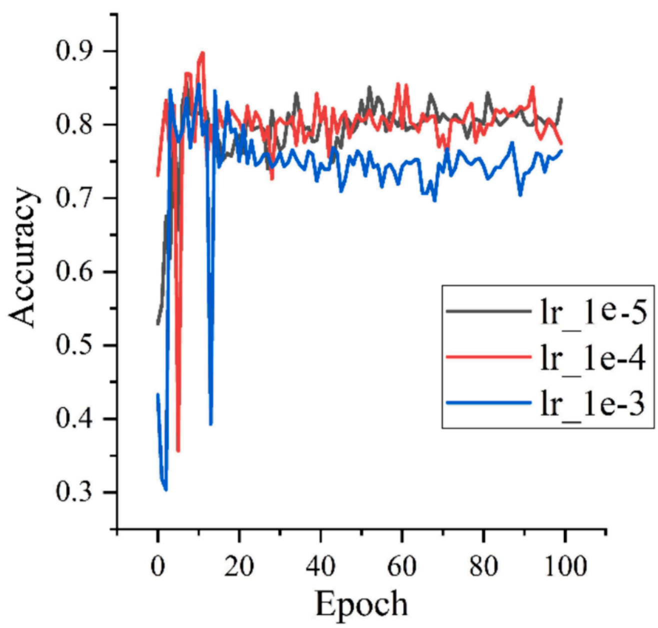

Fig. 12. Accuracy of the model on the validation set with three different learning rates.

In response to the limited capability of network models in extracting time-domain features from time-frequency spectrograms, Yusof et al. [33] classified welding process signals under different penetration conditions using classical time-domain features such as Mean Absolute Deviation, Standard Deviation, and Root Mean Square values. Their findings demonstrated that classical time-domain features can provide the model with learning capabilities on par with those derived from time-frequency spectrograms. In this study, classic time-domain features such as variance, root mean square value, skewness, waveform factor, peak factor, pulse factor, margin factor, clearance factor, maximum value, root amplitude, standard deviation, minimum value, peak, kurtosis, peak-to-peak value, and average energy are extracted for each signal segment. These features are then jointly inputted into a neural network, eliminating the need for the model to extract typical time-domain features through image texture. This approach enables the model to achieve better performance in the weak time-domain feature extraction. To ensure the independence and balance between image features and extracted time-domain features, the input position of the time-domain features is placed after the convolutional process, and the convolutional features are compressed through global pooling.

To address the issue of most channels in the model failing to capture any features, Ding et al. [34] find that the large kernels can improve the model's receptive field, thus, in this work the size of the convolutional kernels is increased from 3 to 5. This allows the kernels to access more pixels during each convolutional process, providing the model with the ability to simultaneously incorporate more feature information. The improved model is depicted in Fig. 14.

In response to the limitations of evaluating the quality of results solely from a mathematical perspective during the training process of convolutional networks, it is essential to ensure that the model not only makes predictions from a mathematical standpoint but also incorporates pre-existing knowledge from the real world. McGowan et al. [35] have achieved higher prediction accuracy of porosity rate in additive manufacturing by including physical meanings in the neural network loss function. To address this, this study introduces a customized loss function that incorporates dual metrics, encompassing both mathematical and physical aspects, in order to incorporate prior physical knowledge. This enables the model to have a more targeted learning direction, considering both mathematical accuracy and adherence to established physical principles.

The loss function L_c is defined as follows:

$$L_c = L_{bce} + \lambda \times L_{feature} \tag{3}$$

$$L_{feature} = -\left(\frac{RMS}{S_{RMS}}\right)^2 \tag{4}$$

where L_{bce} is the cross-entropy loss function used to evaluate the difference between the predicted labels and the true labels, and $L_{feature}$ represents the ratio of the root mean square value of the signal to the relative RMS value, reflecting the degree of energy accumulation in the signal segment. A larger RMS value indicates more energy accumulation, which encourages the model. Therefore, $L_{feature}$ is negative. To balance the L_{bce} loss, a weight coefficient λ is introduced before $L_{feature}$. Based on previous training experience, λ is set to 0.001.

Fig. 15 shows the loss and accuracy of the validation set in the training process. It can be seen that after more than 40 iterations, the loss has basically dropped to near 0, and the accuracy of the model on the validation set has basically stabilized above 96 %, the model has

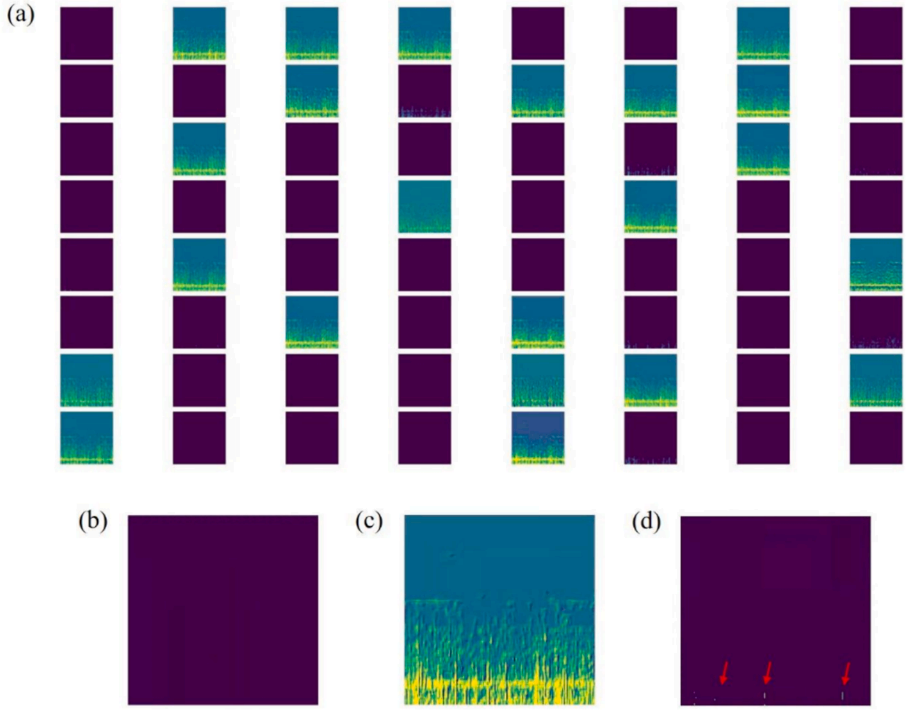

Fig. 13. Visualization results of the convolutional layers in CNN. (a) Overall results, (b) Convolution channels have learned no features, (c) Convolution channels that have learned texture information from the spectrograms, (d) Convolution channels that have learned important time-domain information from the spectrograms.

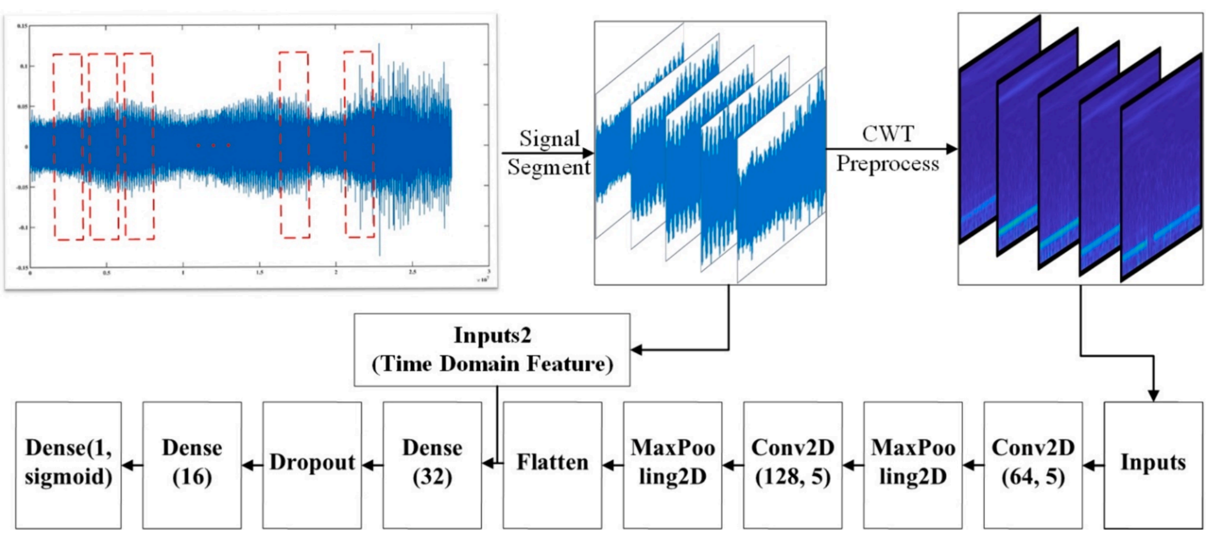

Fig. 14. Structural diagram of the improved model. In addition to pre-processing with CWT, classic time-domain features were extracted from segmented signals to form feature vectors. These feature vectors, along with spectrograms, were jointly input into the neural network to enhance its learning width.

converged. The model takes 0.13 s from data reading to complete prediction.

After implementing the improvements, the model was evaluated using the metrics of Accuracy, Precision, and Recall.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

where TP represents the number of true positives, TN represents the number of true negatives, FP represents the number of false positives, and FN represents the number of false negatives. The Accuracy metric provides an intuitive measure of the model's overall predictive performance, reflecting the proportion of correct predictions. Precision primarily focuses on the number of false positives, and a higher Precision indicates that the model has fewer instances of misclassifying negatives as positives, thereby reducing the risk of false alarms. Recall primarily considers the number of false negatives. A higher Recall implies that the model captures more positive samples, indicating a lower risk of missing positive instances.

Among these metrics, Accuracy is the most straightforward

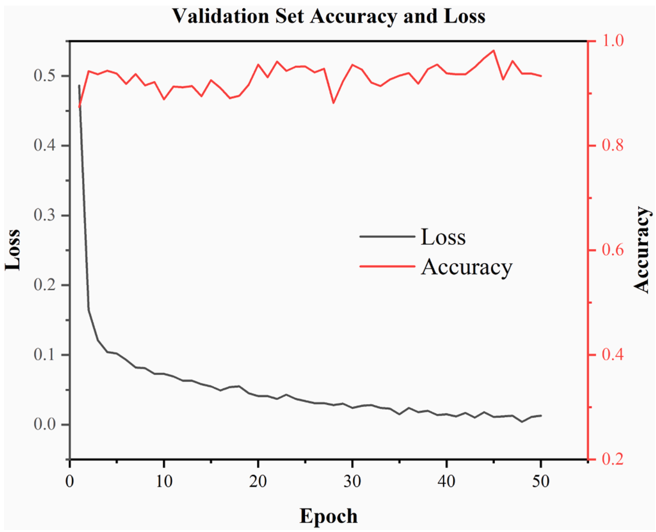

Fig. 15. The accuracy and loss of the validation set in training process.

measurement, representing the proportion of correctly predicted instances and assessing the overall classification accuracy of the model. Precision emphasizes the quantity of false positives, while Recall focuses on the quantity of false negatives. Higher Precision indicates fewer instances of misclassification, while higher Recall suggests capturing more positive instances and minimizing the risk of false negatives.

After incorporating the time-domain features and customizing the loss function, the training results of the model showed significant improvement with the highest accuracy on the validation set reaching 96 %. To verify the generalization ability of the model, we evaluated the performance of the model through a test set consisting of 1776 time–frequency graphs of welding success signals and 1776 time–frequency graphs of welding failure signals and their corresponding acoustic signal characteristics. Table 3 and Fig. 16 illustrate the changes in accuracy, precision, recall on the test set for the model trained with only image inputs and the model trained with the inclusion of time-domain features and the customized loss function.

4. Conclusion

This research monitored the connection process of femtosecond laser glass-copper dissimilar material welding using acoustic emission signals and achieved the prediction of welding success through a neural network. The results demonstrate that the acoustic emission signals can capture characteristic signals during femtosecond laser processing. The specific results of this research are as follows:

(1) A novel feature of "Sum of Relative RMS" was defined to characterize the energy accumulation in the repeated welding process, addressing the impact caused by minor surface disturbances on the variation of initial welding signals. The correlation between the formation of connections and acoustic emission signals was established by the value of "Sum of Relative RMS" and increase in signal amplitude, allowing for evaluation of the quality of femtosecond laser glass-copper welding, providing a reference for monitoring welding quality using acoustic emission signals.

(2) This study combines the spectrogram generated by the Discrete Wavelet Transform of acoustic emission signals and conventional time-domain features. By incorporating "Sum of Relative RMS" features and a customized loss function based on prior physical knowledge, the performance of the Convolutional Neural Network model is improved. Currently, the accuracy on the validation set has been increased from 89 % to 95 %, demonstrating sufficient reliability and accuracy for industrial applications after multiple predictions.

Table 3
Evaluation metrics for the two training methods.

Method	Accuracy	Precision	Recall
Before improvement	0.89	0.862	0.930
After improvement	0.95	0.923	0.988

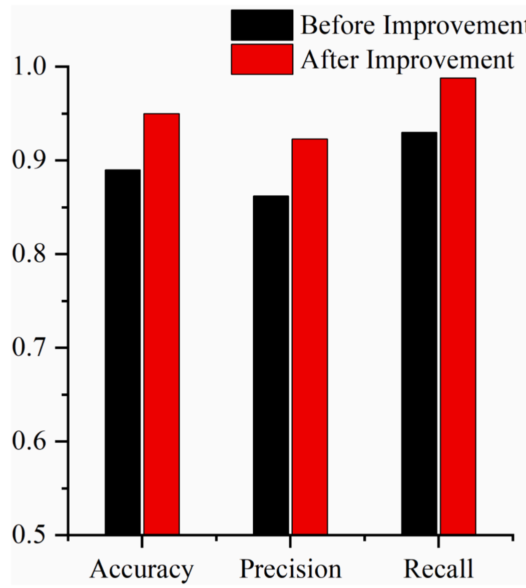

Fig. 16. Comparison of each index of the two training methods.

CRediT authorship contribution statement

Wei Wei: Writing – review & editing, Conceptualization. **Yang Liu:** Writing – review & editing, Writing – original draft, Methodology. **Jindou Wu:** Methodology. **Zhilin Wei:** Software. **Zhukun Zhou:** Writing – review & editing, Supervision. **Yu Long:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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